

Federated Chiplet Design and Simulation

An OCE/CDX Framework for Multi-Vendor Co-Design Without IP Disclosure

David Ratchkov

Kevin Cameron

What This Talk Covers

- The multi-vendor chiplet integration problem
- A guiding principle: separate model ownership from model execution
- CDXML, the Simulation Capability Manifest, and a federated architecture
- Multiphysics co-simulation, and IP protection as part of the standard
- How this fits with existing OCE virtual prototyping and Accellera FSS
- AI-driven design-space exploration and a phased proof of concept

Why Chiplet Co-Design Is Hard

- Before selecting chiplets for a System-in-Package, an integrator must know whether chiplets from different suppliers can operate together as a system — which requires analysis across:
  - Architectural and functional interoperability
  - Signal integrity
  - Power integrity and transient power behavior
  - Thermal interaction
  - Mechanical stress and warpage
  - Package and assembly constraints
  - Workload-dependent performance and power
  - Reliability and lifecycle behavior (eventually)

The Structural Problem

- The models required for these analyses often contain proprietary information — floorplan, power distribution, implementation detail, process information, RTL, SerDes behavior, and other differentiating IP.

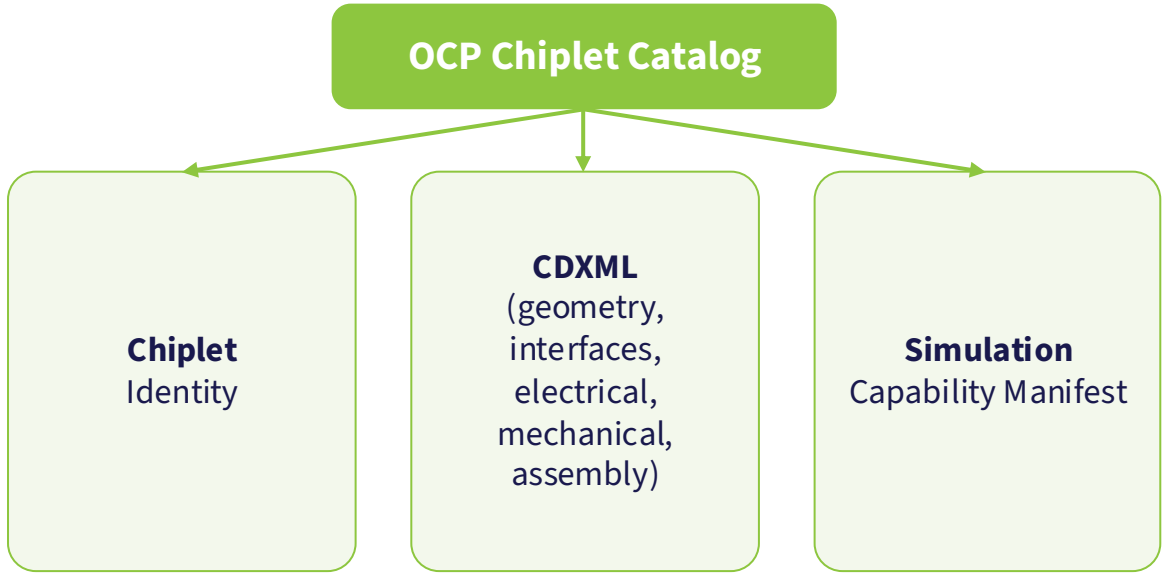
"How can an integrator co-design a system containing chiplets from multiple companies when none of those companies wants to expose the detailed models required for system-level simulation?"

Separate Ownership From Execution

- Instead of requiring a chiplet vendor to hand over its proprietary simulation model, the vendor exposes a standardized simulation service
- The integrator submits permitted boundary conditions and receives permitted simulation results
- The proprietary model remains under the chiplet supplier's control
- **In simple terms: bring the simulation request to the model, rather than bringing the model to the integrator**
- This creates a form of Model-as-a-Service for chiplets

Building on CDXML

- CDXML already provides a machine-readable description of chiplet information for design, integration, assembly, mechanical, I/O and electrical purposes — created to automate design and business processes across company boundaries
- Federated simulation builds on this foundation: a CDXML chiplet description can reference an associated Simulation Capability Manifest describing what simulations the supplier makes available
- CDXML describes both what the component is, and enough information to discover what analyses can be performed against it — the detailed simulation implementation does not have to reside in CDXML



This allows discovery to occur automatically.

What a Simulation Capability Manifest Describes

Identity & Version
Chiplet, vendor, model ID and version, model hash, applicable revision

Simulation Domain
Thermal, power, PI, SI, mechanical, functional/architectural, performance, reliability

Boundary Contract
Permissible inputs/outputs, units, ports, operating ranges, timestep requirements

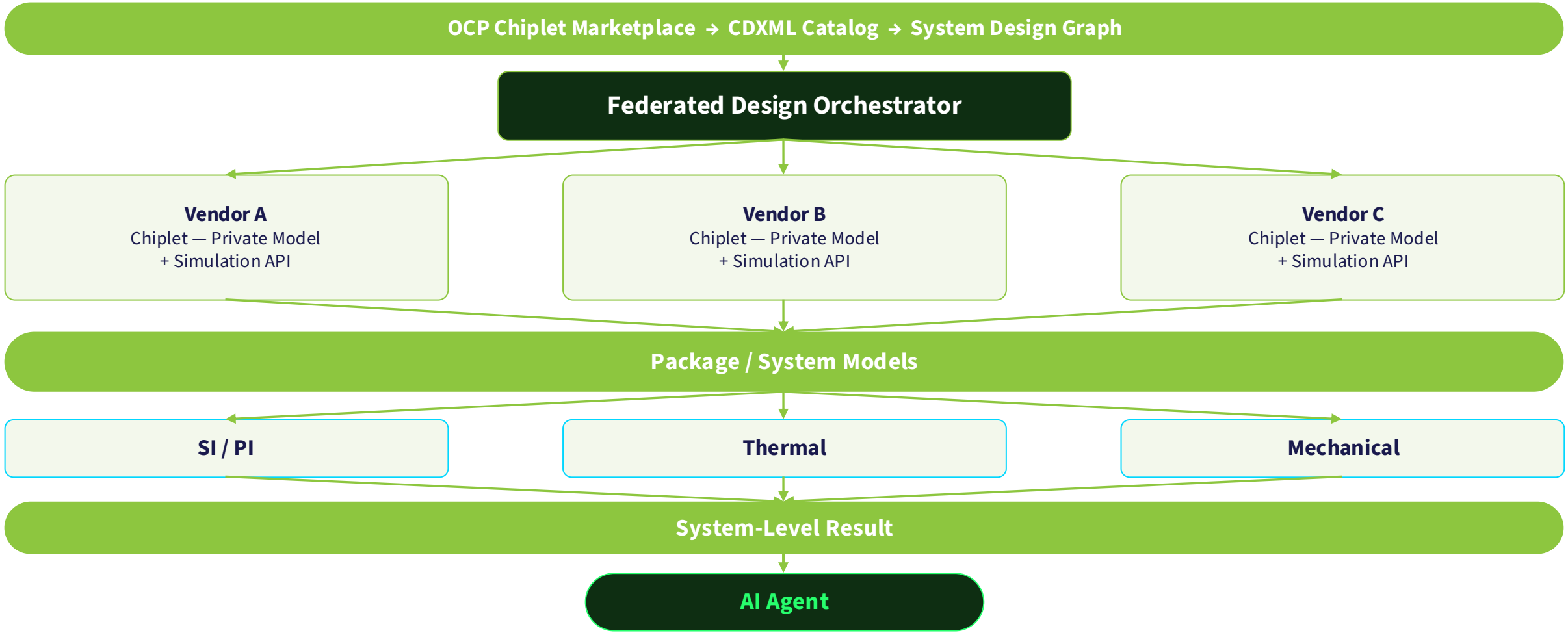
Model Validity
Fidelity class, validated range, assumptions, error bounds, prohibited uses

Execution Method
Vendor-hosted service, encrypted/containerized model, confidential computing, reduced-order model

Security & Authorization
Authentication, entitlement, attestation, result granularity, query limits, audit

Commercial Policy
Open, licensed, NDA required, per-query, subscription, or evaluation-only

Federated Simulation Architecture



Multiphysics Co-Simulation

1. A workload generates activity requirements
2. The chiplet performance model translates activity into a power profile
3. A package PI solver calculates supply behavior
4. Chiplet models respond to voltage conditions
5. Thermal models calculate temperature
6. Temperature modifies leakage and performance
7. Mechanical/package models determine additional constraints
8. The system iterates until convergence

Only the agreed boundary values need to cross organizational boundaries — the models can reside at different companies.

IP Protection Is Part of the Standard

Federation alone doesn't guarantee confidentiality — a malicious integrator could run many carefully chosen simulations to reverse-engineer a model. The framework needs an explicit confidentiality policy supporting:

- Authenticated users
- Authorization policies
- Encrypted communications
- Signed model identities
- Model / version attestation
- Query logging
- Rate and query limits
- Restrictions on observable outputs
- Minimum result aggregation
- Protection against model extraction
- Secure execution environments where appropriate
- Audit trails

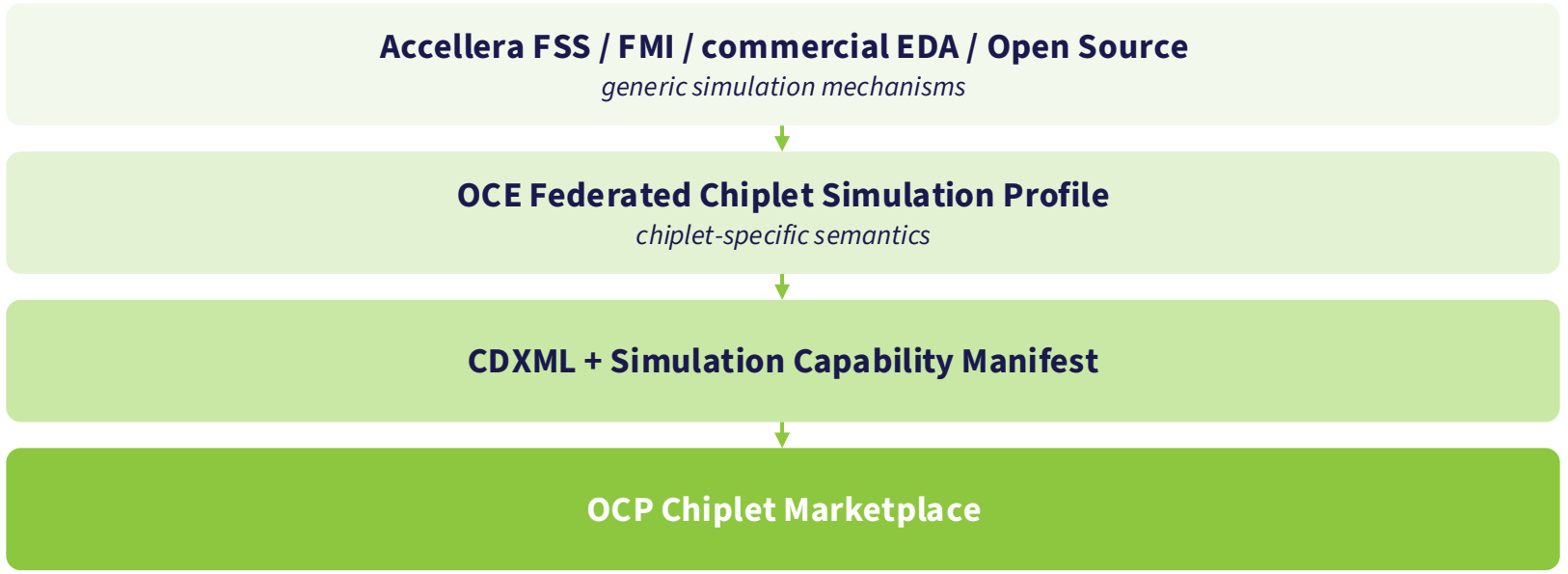
OCP already has active work on device identity and attestation — this framework should reuse those concepts rather than inventing a new trust architecture.

Extending OCE Virtual Prototyping

- This framework is positioned as an extension of the existing OCE virtual-chiplet work
- The current OCE proposal includes a centralized repository of SystemC-like models, private repositories, access controls, simulation/emulation/prototyping, and automated design-space exploration
- The July 2026 OCE direction already adds AI automation, encrypted/containerized models, cloud execution, and local or remote verification IP
- **Federated Chiplet Simulation adds a stronger requirement: a supplier must be able to participate in system-level simulation without transferring its proprietary model to either the integrator or a central repository**
- The existing repository approach remains valuable for models vendors are comfortable sharing — federation becomes another execution class

Aligning With Standards

- OCE should not create a generic distributed-simulation protocol unless it is needed
- Accellera's Federated Simulation Standard (FSS) promises to address interoperability among distributed modeling and simulation environments, and connects existing modeling standards and tools
- **OCE/CDX should define what a chiplet simulation service means, rather than reinventing how simulators communicate**



Federated Simulation Implementation

- Individual simulations run in secure containers to protect IP
- Wire, RF and photonic connections become socket connections between the containers
- Leverage VLAN and DNS features of OCP server environments
- Socket connections can be re-plumbed to go through wiring models
- Containers are hot-swappable/pluggable for fast design exploration
- Simulations for environmental concerns (thermal, noise) are pluggable for design refinement
- Specialist tools handle thermal modeling and Chiplet networking
- Use open-source reference implementations (vs standards committees, SV/VHDL/Verilog-AMS based)
- **OCE/CDX defines**
  - **connection protocols and resource allocation for simulators (SMP/GPUs/FPGA/HiL)**
  - **mechanisms for locating repositories of components, and normalized search**
  - **tool registration (simulators, accelerators, connectivity synthesis)**
  - **bridges to exiting tool standards (SV/DPI, SystemC-TLM, FSS)**

AI Sits Above Trusted Simulation

AI should initially sit above trusted simulation — not replace it. A design-space exploration agent could:

- Understand system requirements and query the OCP chiplet catalog for candidate components
- Construct candidate SiP architectures and discover simulation capabilities automatically
- Execute federated simulations and reject infeasible combinations
- Learn surrogate relationships from permitted results and choose subsequent experiments intelligently
- Rank architectures by PPAC and other objectives

"Find a five-chiplet architecture providing at least 800 GB/s memory bandwidth, staying below 600 W package power and 90°C junction temperature while minimizing package cost."

The agent can evaluate many combinations while never obtaining the proprietary models behind the components.

A Deliberately Simple Proof of Concept

Phase 1 — Initial POC

- Two independently modeled chiplets, one substrate/package, CDXML for both, two simulation servers (two suppliers), one federated orchestrator
- Physics: power + thermal only — not yet SI, PI, mechanics or functional simulation
- Integrator specifies placement, package characteristics, cooling, workload, ambient conditions
- Output: junction temperatures, package power, thermal coupling, performance-limit violations, feasible/infeasible classification
- Then: automated design-space exploration moves chiplets / changes workloads while both private models stay inaccessible

Phase 2 — Adds

- Transient PI
- Workload/performance model
- Functional SystemC model
- Multiple chiplet choices from the catalog
- **The AI agent can then select the actual combination of chiplets — not just placement**

Five Concrete Deliverables (Plus One)

The project does not initially need another large specification — five concrete artifacts are sufficient:

1. **Federated Chiplet Simulation Use Cases & Requirements**
2. **Chiplet Simulation Capability Manifest**
3. **Federated Chiplet Simulation Interface / Profile**
4. **Security, Trust and Model-Confidentiality Requirements**
5. **Open Reference Demonstrator**

A sixth deliverable can eventually be a Federated Simulation Conformance Test Suite, so chiplet vendors and EDA providers can claim compatibility.

From a Catalog to a Design Environment

- **The OCP Chiplet Marketplace evolves from a catalog of components to a design environment for components.**
- Chiplet suppliers need not expose their designs to the integrator — or to each other.
- **That ability would remove one of the most fundamental barriers to a true multi-vendor Open Chiplet Economy.**

“Add these components to a virtual SiP and determine whether this architecture satisfies my power, thermal, interconnect, mechanical and performance requirements.”

Thank You!

Click here to review OCP Trademark Guidelines



Open Discussion

Click here to review OCP Trademark Guidelines